Lists in python:-

A list in Python is a collection of items stored in a single variable. Lists are ordered, changeable (mutable), and allow duplicate values.

List of rooms with Air conditioning facility and without the facility :-

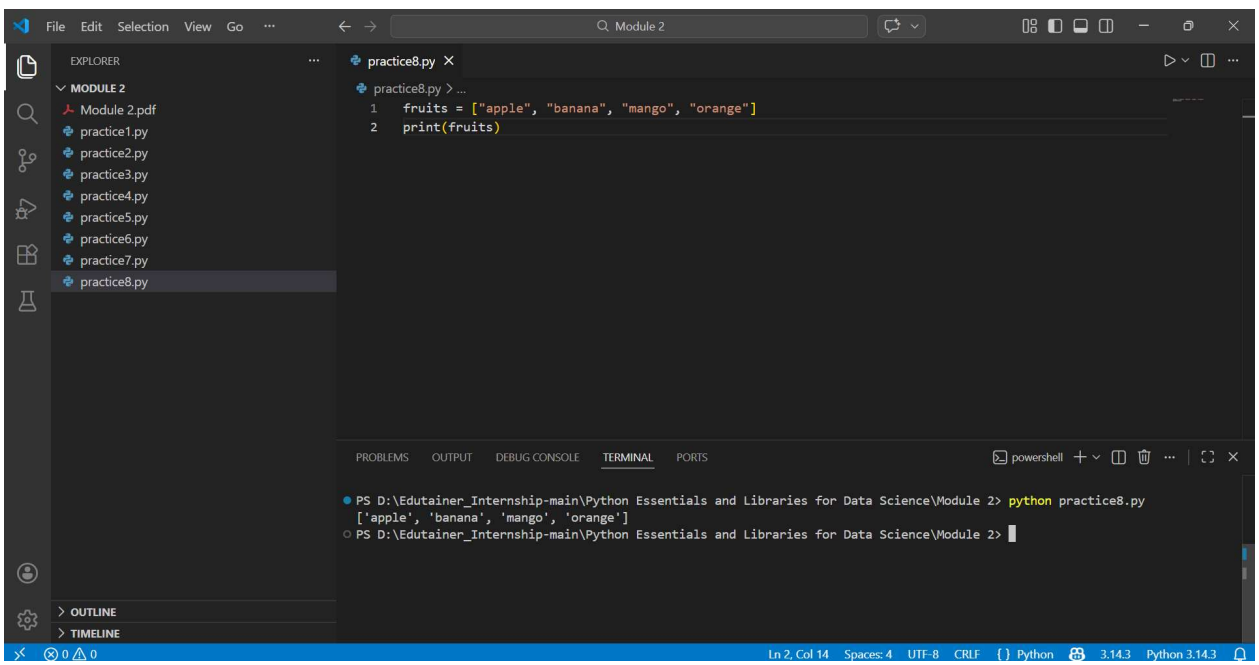
Ac_rooms = [2,6,7,8,9]
Non_ac_rooms = [1,4,10,12,13]

Items ordered in last 7 days:-

Dish_ordered = ["idli", "dosa", "dosa", "pulav", "dosa", "vada", "idli"]
Hostel_items = ["clothes", "washing powder", "mouse"]

Creating a list in python:-

Lists are created using square brackets [ ].

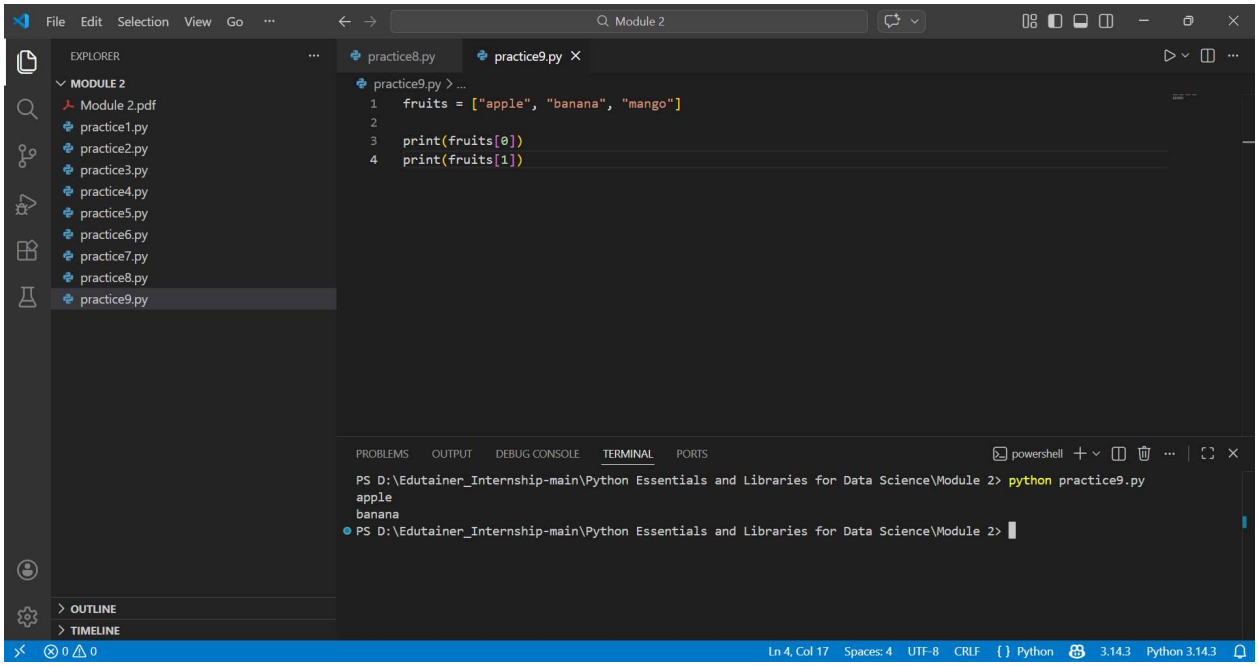


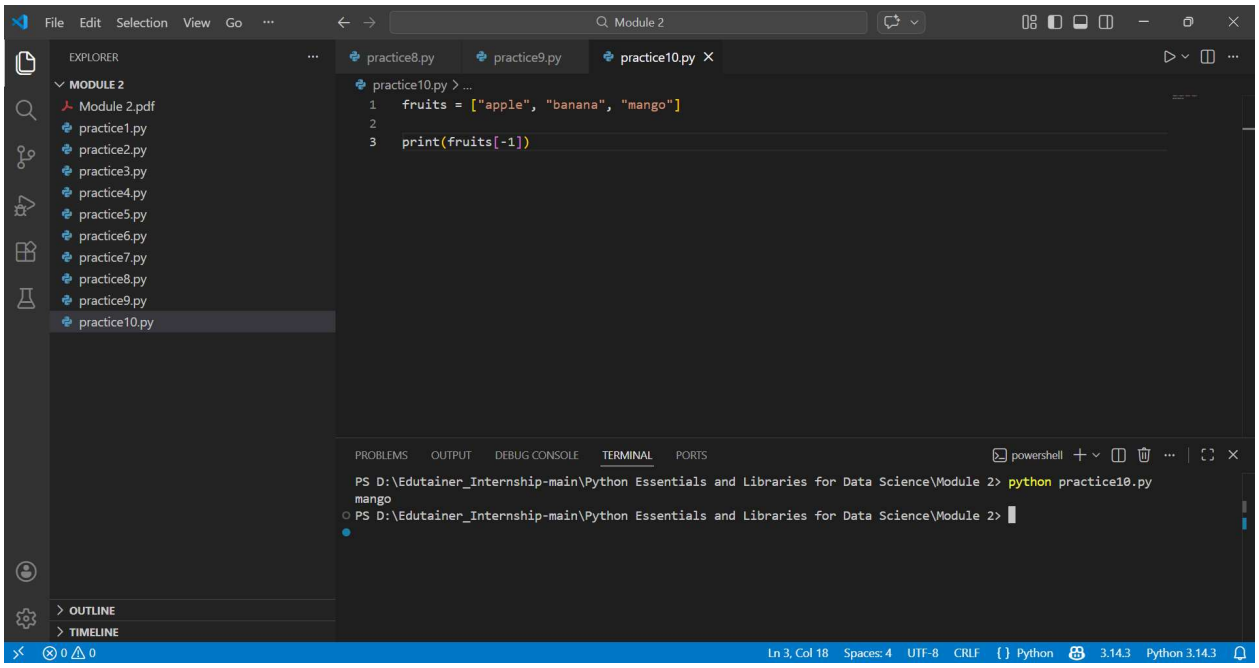
Characteristics of lists:-

Ordered : Items have a defined order.
Mutable : Items can be changed after creation.
Allow Duplicates : Same values can appear multiple times.
Different Data Types : A list can contain multiple data types.

Accessing list items:-

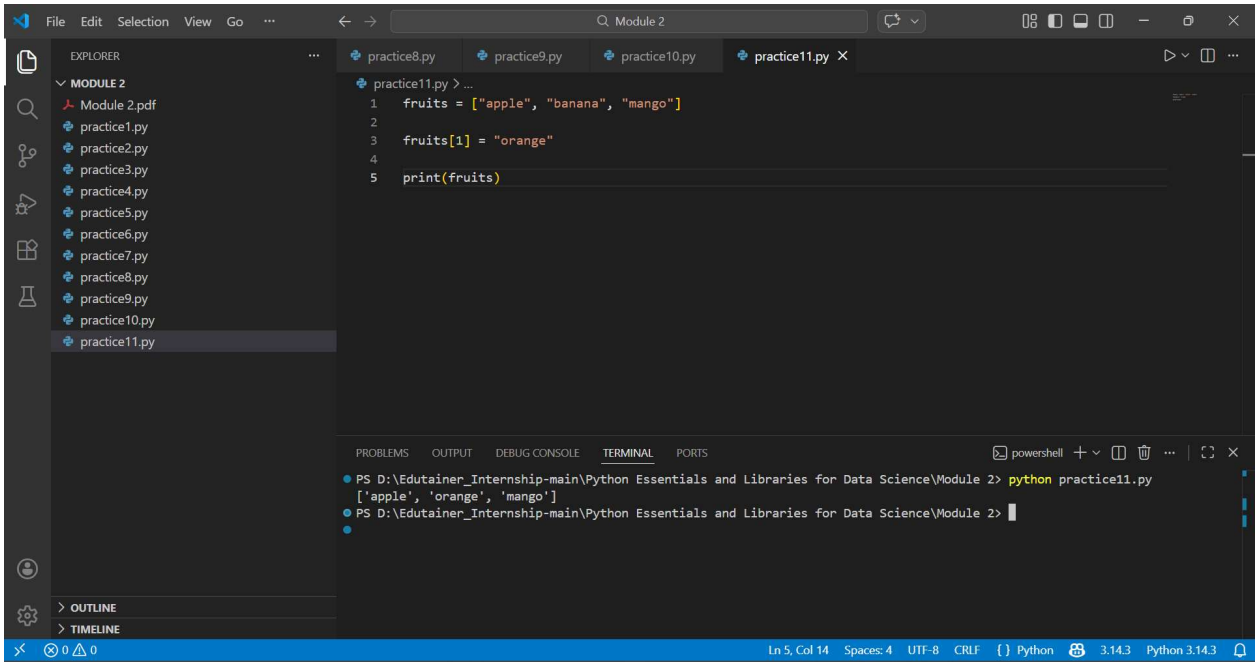
List elements are accessed using **index numbers.**Negative index starts from the **end of the list.**





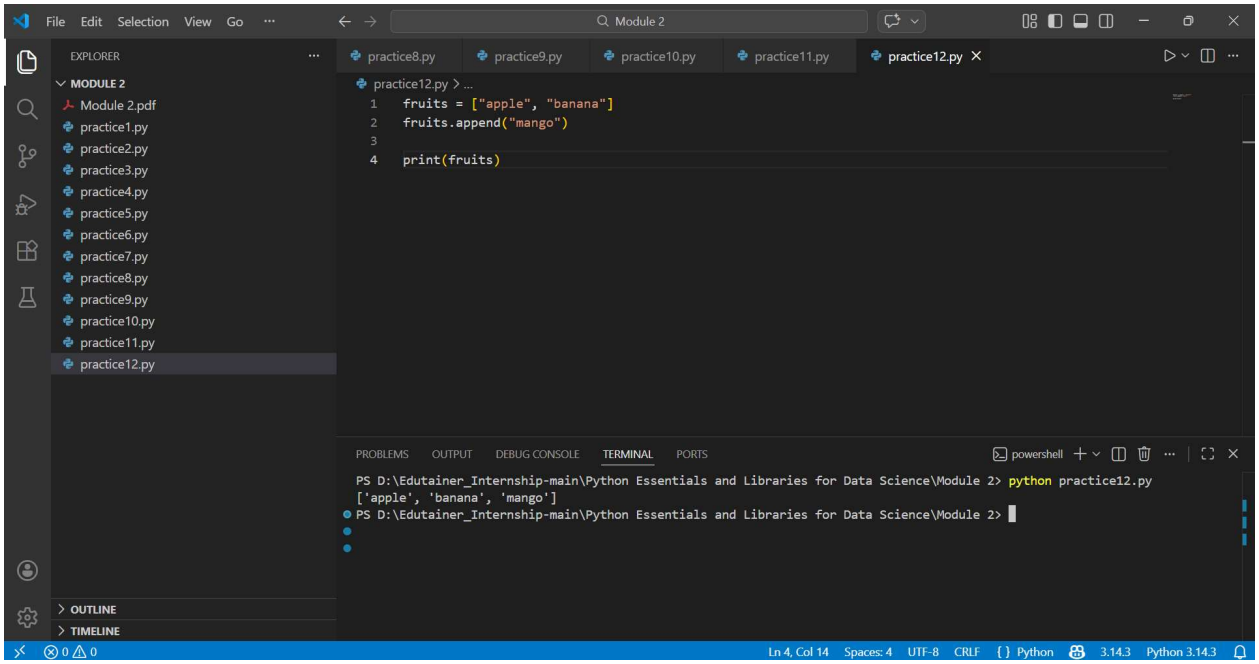
Changing list elements:-

List elements can be change using index.

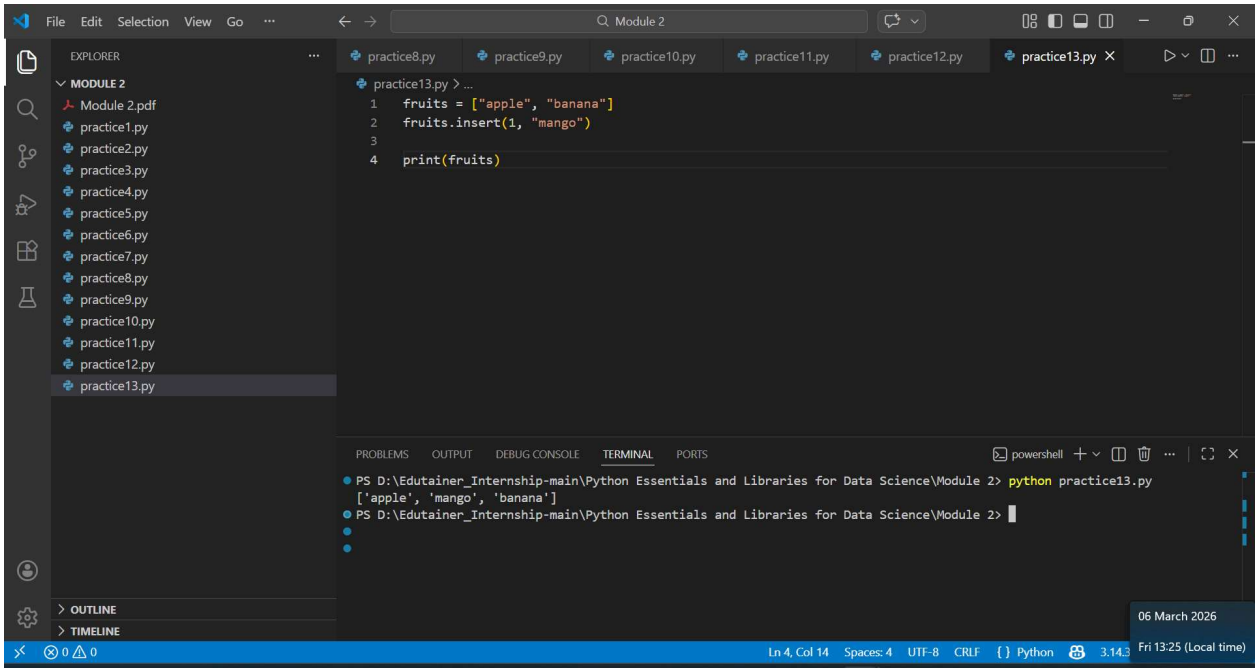


Adding elements in the list:-

append() : Adds element at the end

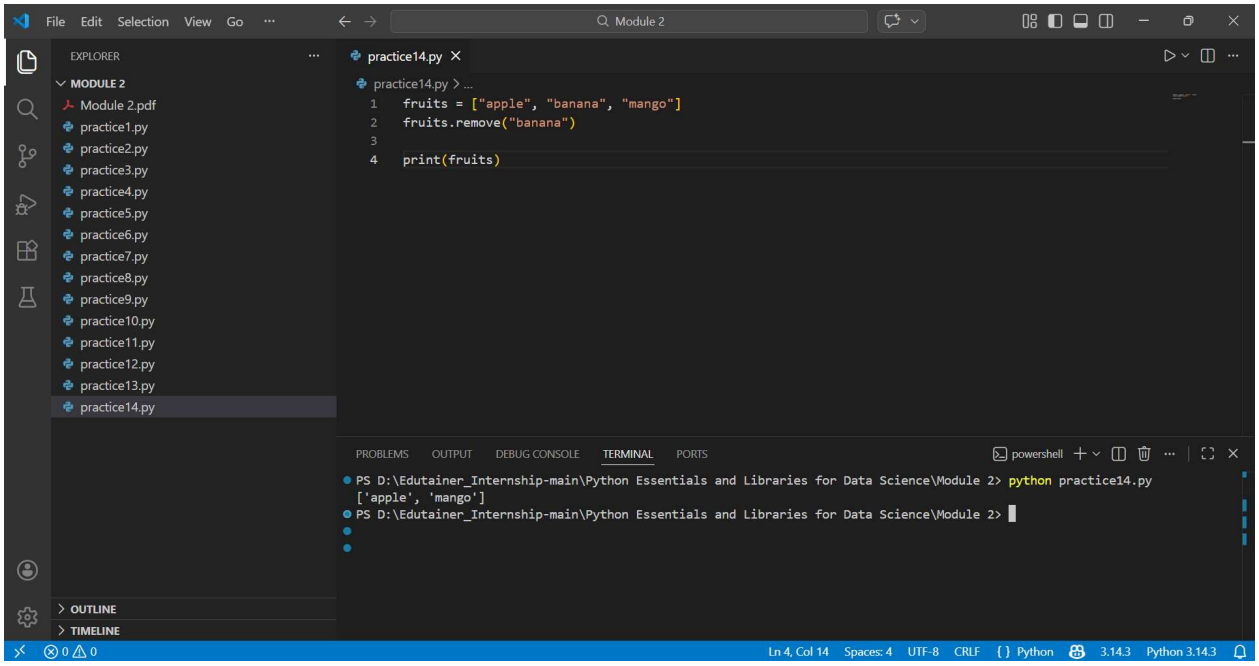


insert() : adds element at specific position.

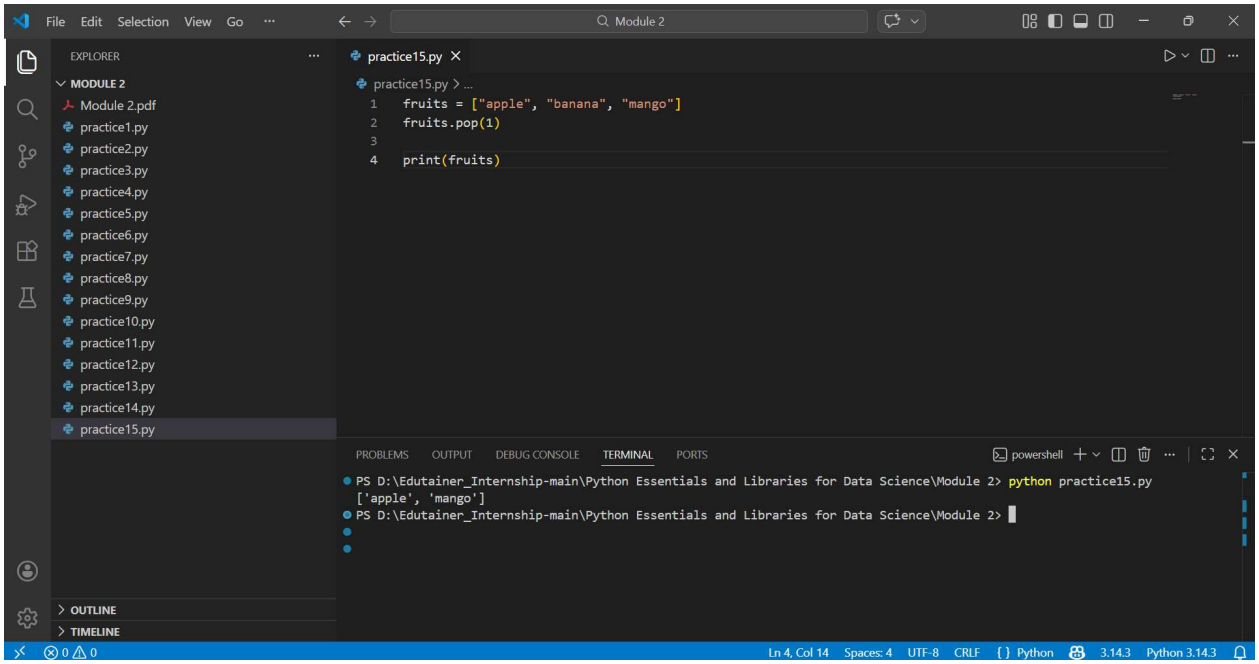


Remove elements from the list:-

remove() : removes specific value

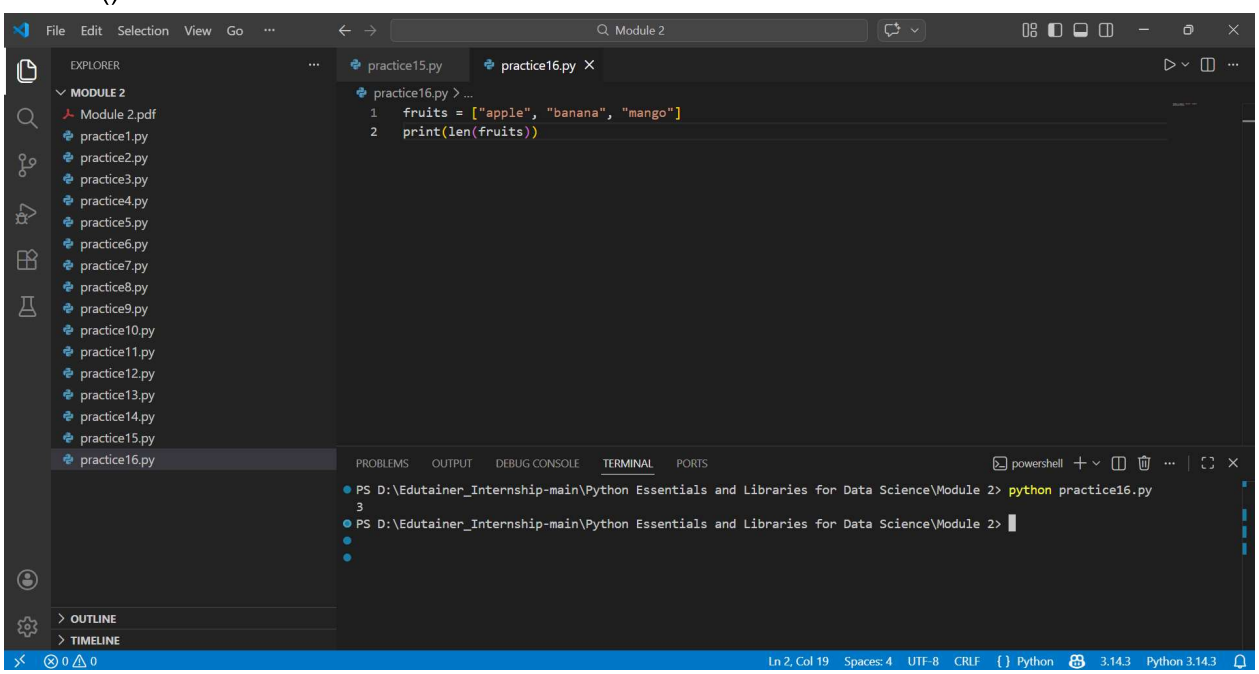


pop() : remove elements using index



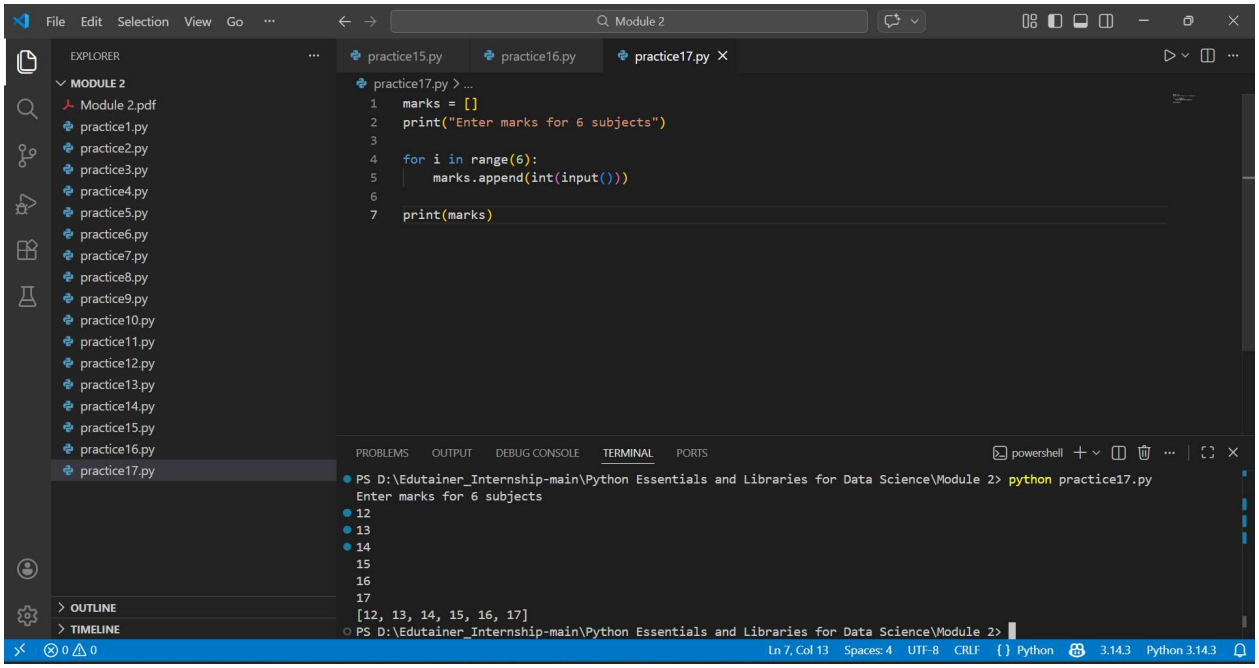
Length of list:-

Use len() to find number of elements.

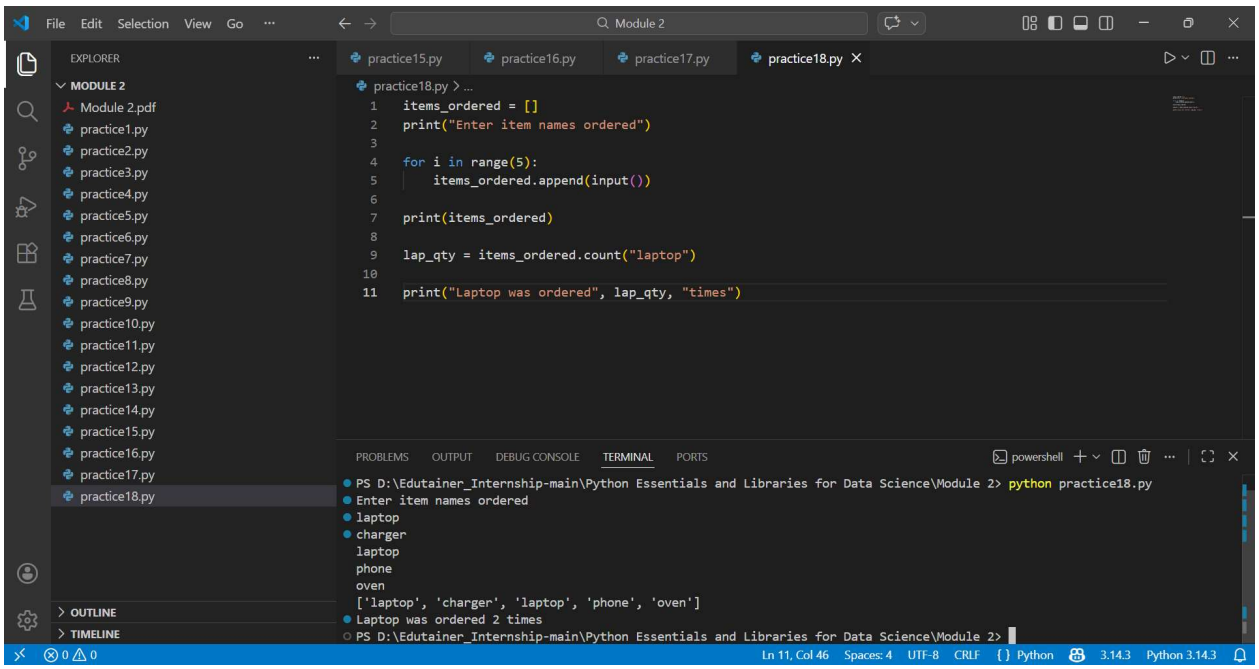


Example programs:-

1.Print the marks of 6 subject.

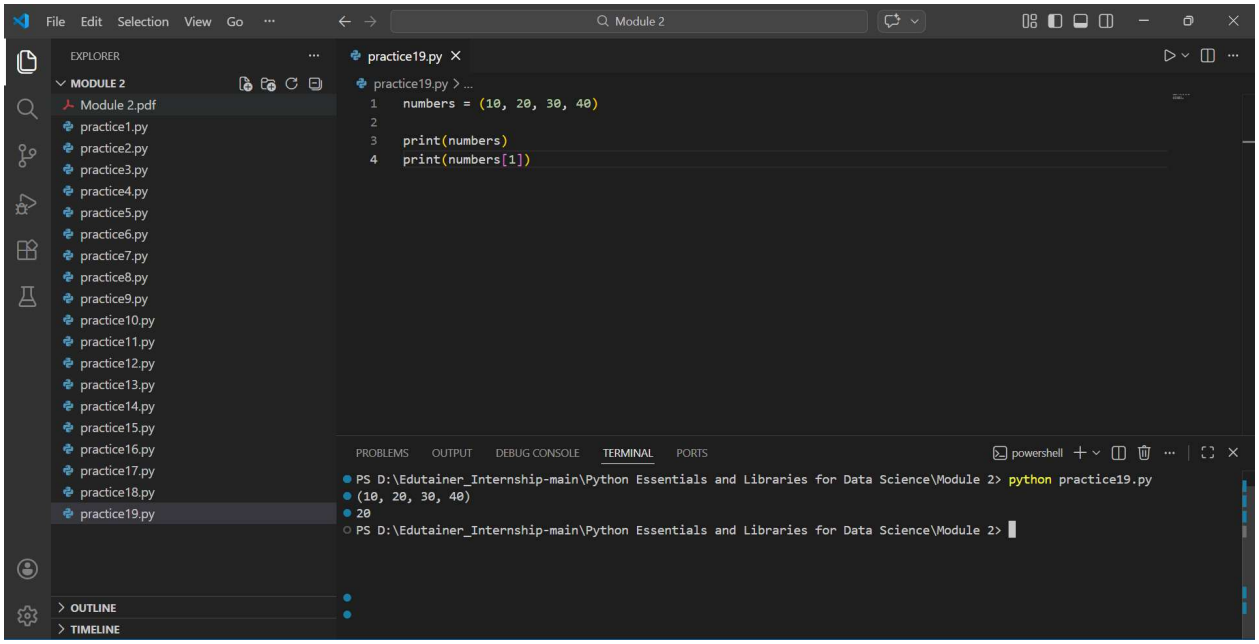


2.Count the number of times the laptop was ordered.



Tuple in python:-

A tuple is a collection of elements that is ordered and immutable (cannot be changed).

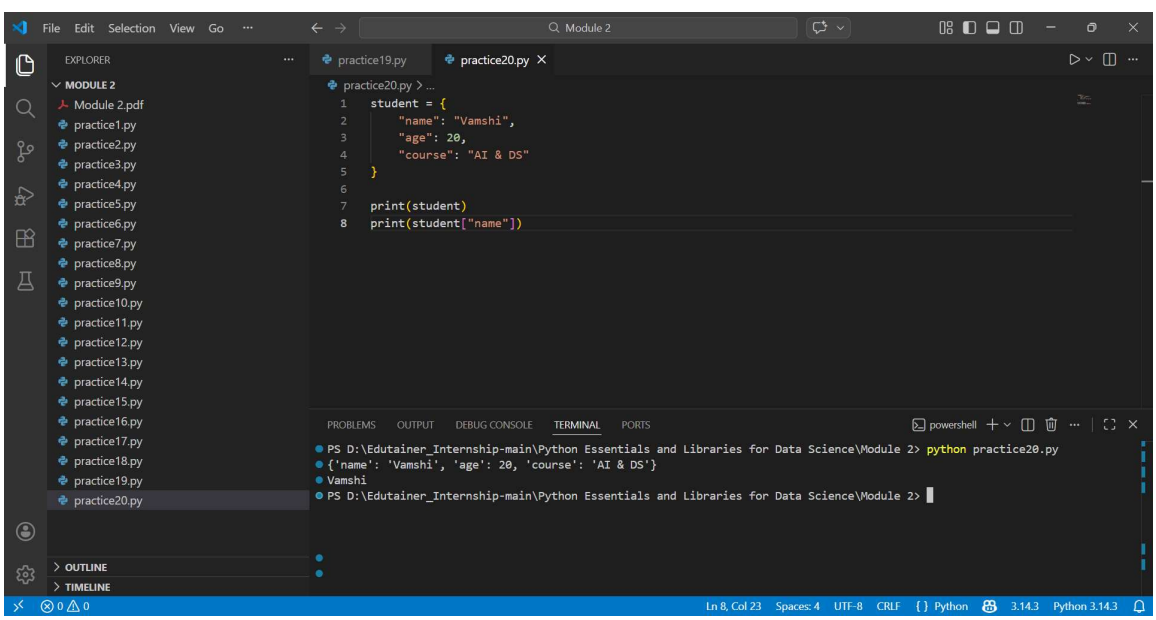


Characteristics of tuple:-

Ordered
Immutable (cannot modify values)
Allows duplicate values
Written using ( ) parentheses

Dictionary in python:-

A dictionary stores data in key–value pairs.

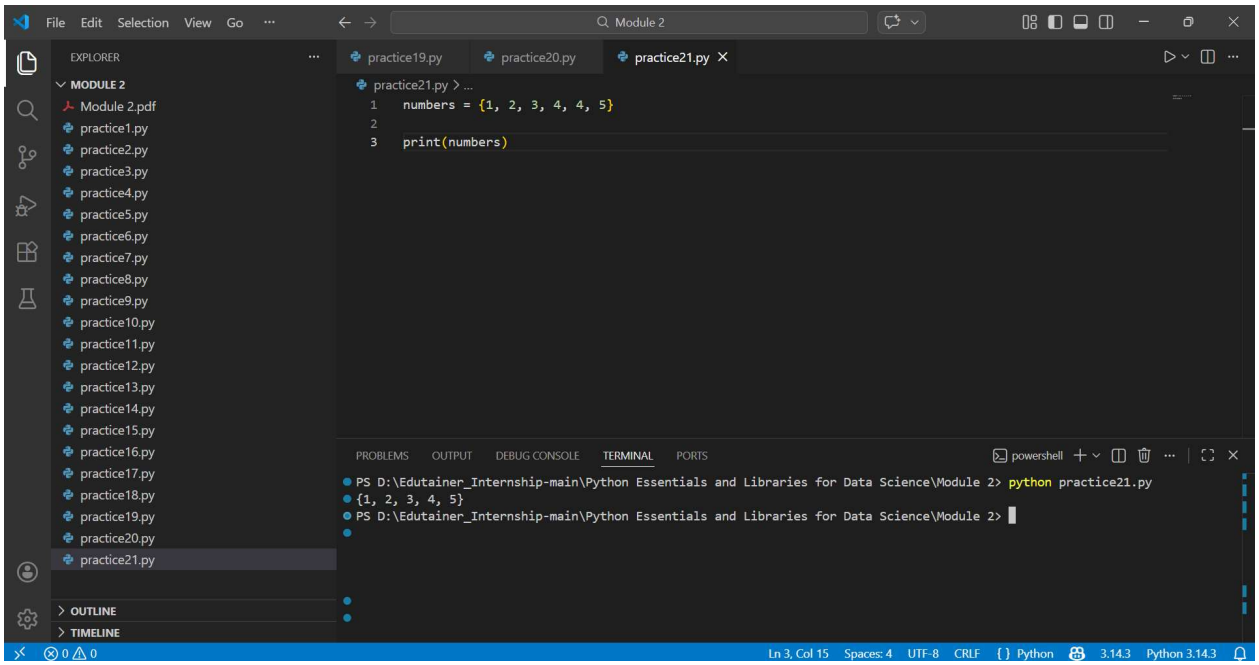


Characteristics of Dictionary:-

Stores key-value pairs
Mutable (values can be changed)
Keys must be unique
Written using { }

Set in python:-

A set is a collection of unique elements. Duplicate values are not allowed.



Characteristics of set:-

Unordered
No duplicate values
Mutable
Written using { }